# Poster Grid Recognition & Retrieval

# 海报网格识别与检索工具

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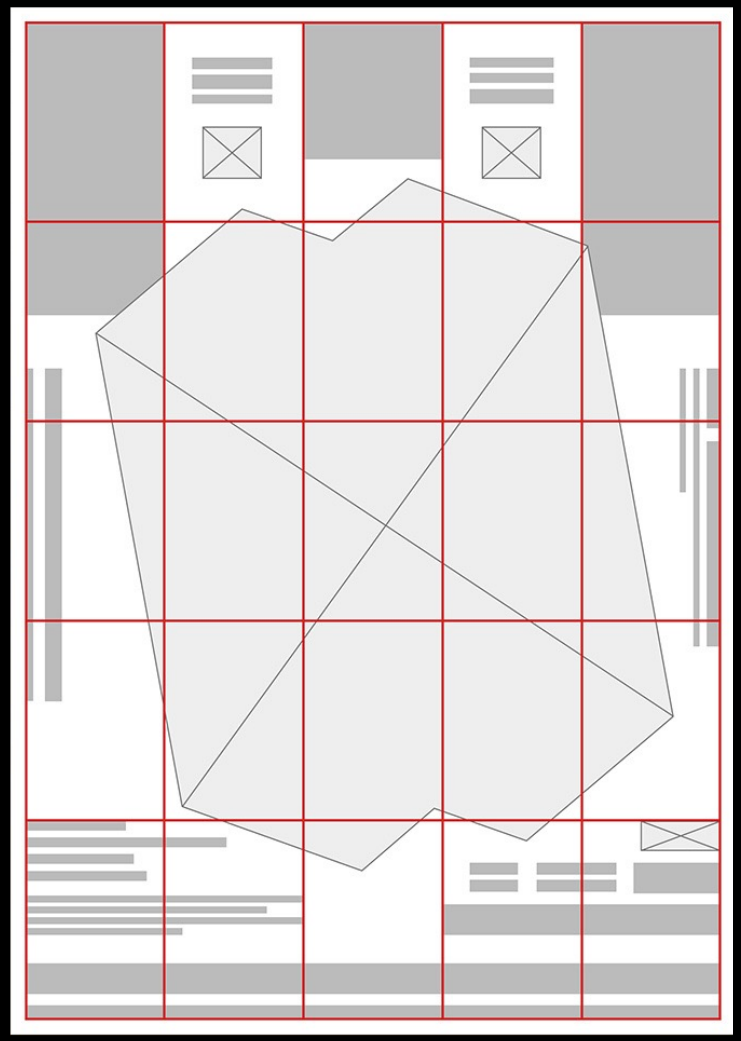
## Design Purpose

## 设计目的



In this semester's design project course, the study and research of poster grid systems is of great importance. In this course, we learned about and investigated the grid systems used in International Style posters, which made us realize the significance of grid systems in graphic design. For this reason, we intend to design a webpage that actively annotates, indexes, and analyzes posters, to provide support for our design practice.

本学期的设计项目课中，海报网格系统的学习和研究十分重要。在这门课上我们学习与研究国际主义海报的网格系统，这让我们意识到网格系统在平面设计中的重要性，因此我们想设计一款主动标注、索引、分析海报的网页，为我们的设计实践提供帮助。

## Design Style

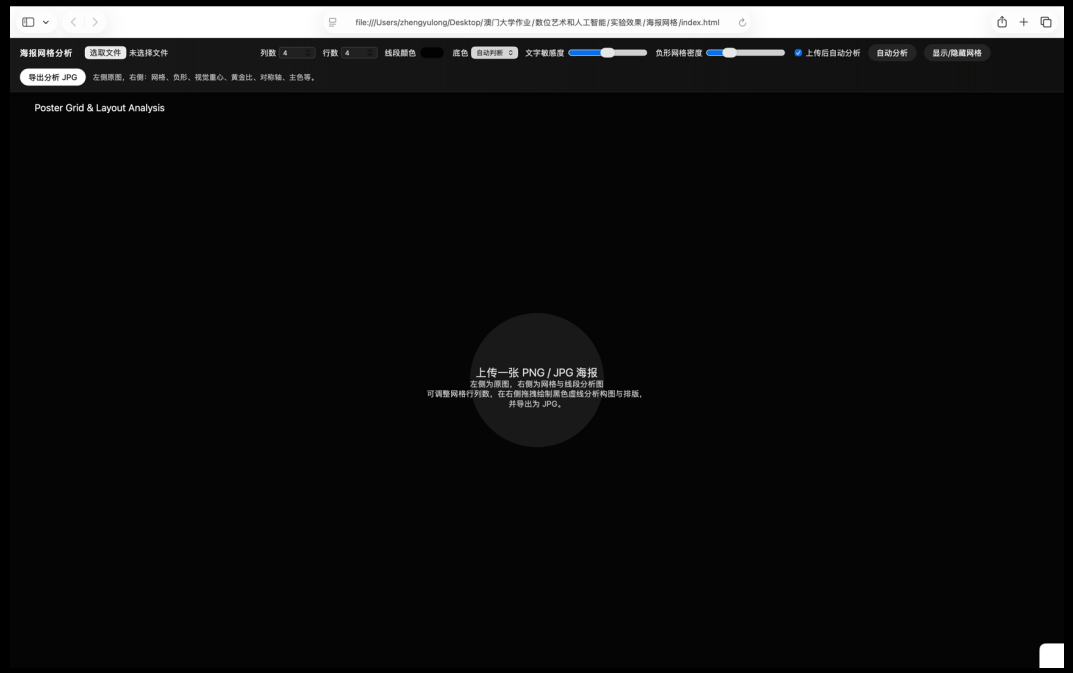
## 设计风格

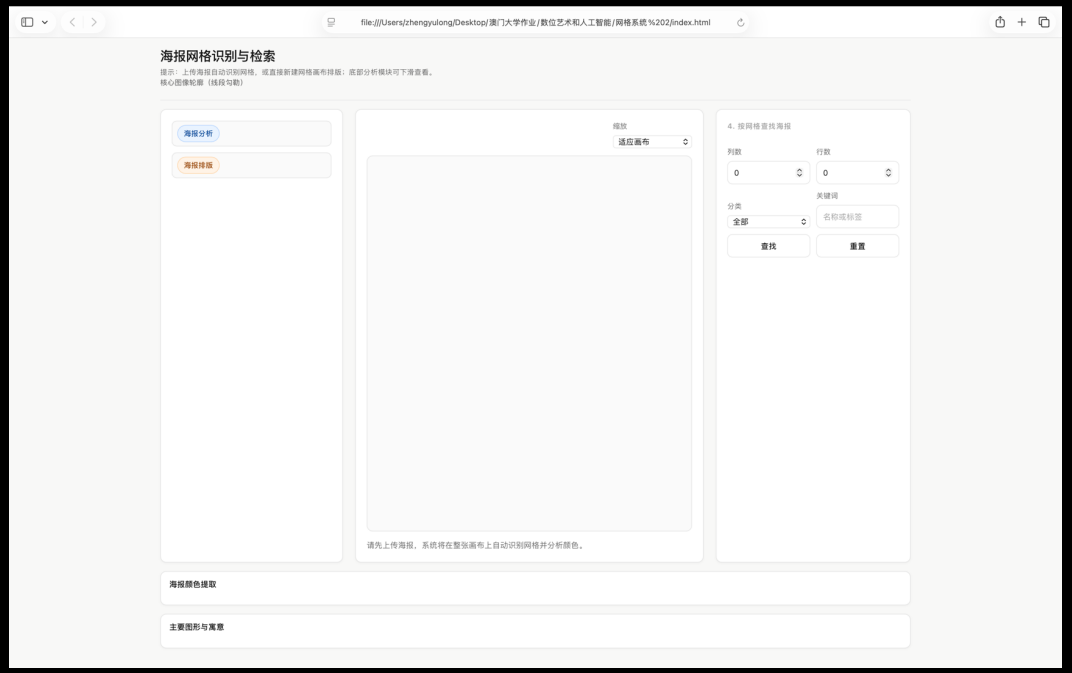
At first, we each designed two styles: black and white. Later, in practice, we found that the white webpage was more efficient to operate, so we focused on the white style for further development.

一开始我们两人分别设计了两种风格，黑色与白色。后来在实际操作中，我们发现白色的网页操作更加高效，我们便以白色为主进行后续深化。

**Concise / Clear / Easy to operate**

简洁 / 清晰 /易操作



## Design description

设计说明

| 按键 | 作用 |
| --- | --- |
| Ctrl/⌘ + Z | 撤销排版 |
| Ctrl/⌘ + Shift + Z | 重做排版 |
| Ctrl/⌘ + Y | 重做排版 |
| ↑ ↓ ← → | 微移当前选中的排版元素（配合 Shift 步长加大，以程序为准） |
| Delete / Backspace | 删除当前选中的排版元素 |
| Esc | **清空画布/海报**（有内容时，见第 0 节警告） |
| Ctrl/⌘ + O | 打开「选择海报」文件框 |
| Ctrl + V | 粘贴剪贴板图像为海报（全局粘贴监听） |

For ease of use, the webpage can be operated using keyboard shortcuts.

为方便操作，网页可以根据快捷键进行操作。

# Design description

## 设计说明

### Analysis · Upload and Recognition
### 分析 · 上传与识别

1. Click "Select Poster (JPG/PNG)" and choose an image from the system file box. Alternatively, drag and drop the file into the central canvas area (stage).Or, copy the image from other software and paste it on this page using Ctrl+V (⌘V on Mac).

2. After successful loading, the system will automatically recognize the grid across the entire image; the central canvas will display the poster and the cyan grid lines.

3. If the effect is not ideal, click "Re-recognize Grid" to recalculate (this button is only available after the image has been loaded).

4. Click "Clear" to clear the current poster, grid confirmation status, color palette, layout analysis, and typography elements, returning to the initial prompt.

1.点击 「选择海报（JPG/PNG）」，在系统文件框中选一张图。或将文件拖放到中间画布区域（舞台）。
或在其他软件复制图片后，在本页按 Ctrl+V（Mac 为 ⌘V）粘贴。
2.载入成功后，系统会在整张图范围内自动识别网格；中间画布显示海报与青色网格线。
3.若效果不理想，点击 「重新识别网格」 再算一遍（需已载入图片后按钮才可用）。
4.点击 「清空」 清除当前海报、网格确认状态、颜色板、版面分析、排版元素等，回到初始提示。

# Design description

## 设计说明

### Analysis: Grid Style and Confirmation
### 分析 · 网格样式与确认

1. Modify the values in the column/row/margin/angle input boxes (column and row ranges are generally 2–36, subject to the interface).

2. Adjust the grid color, transparency, dashed line length, dashed line spacing, and line width to make the grid visible on dark/light-colored posters.

3. Click "Apply Grid Style" to apply the above values to the current canvas and redraw the grid.

4. After modifying columns or rows, you can click "By Canvas Fit": This will indent the grid by the current column and row to the entire canvas, making the cells square and fitting perfectly to the canvas (margins may become symmetrical left and right).

5. Read the text in the gridMeta area to confirm if the current mode is "Indented Canvas," etc.

6. Once satisfied, click "Confirm Current Grid." Only after confirmation can subsequent operations such as "Save to Grid Library" that depend on the "finalized grid" be performed.

1.在列数 / 行数 / 四边距 / 倾角 输入框中修改数值（列、行范围一般为 2–36，以界面为准）。
2.调整 网格颜色、透明度、虚线长度、虚线间隔、线宽，便于在深色/浅色海报上看清网格。
3.点击 「应用网格样式」，将上述数值应用到当前画布并重绘网格。
4.修改列或行后，可点击 「按画布贴合比例」：在整张画布内按当前列:行做最大内接，使单元格为正方形并与画幅贴合（边距可能变为上下或左右对称留白）。
5.阅读 gridMeta 区域文字，确认当前是否为「内接画布」等模式。
6.满意后点击 「确认当前网格」。只有确认后，才能执行后续的「保存到网格库」等依赖「已定稿网格」的操作。

# Design description

## 设计说明

### Analysis & Template Learning
### 分析 · 模板学习

1. After completing grid recognition or adjustment, enter an easy-to-remember name (e.g., "Brand Vertical Main Visual") in the "Template Name" input box.

2. Click "Learn Current Grid" to add the current grid and aspect ratio to the local template list (this button is enabled when conditions are met).

3. Select a template from the "Learned Templates" dropdown menu.

4. To apply the template to the current image, click "Apply Selected Template".

5. To remove a template, select it and click "Delete Selected Template".

1.在完成网格识别或调整后，在 「模板名称」 输入框填写易记名称（如「品牌竖版主视觉」）。
2.点击 「学习当前网格」，将当前网格与画幅比例记入本机模板列表（按钮在条件满足时启用）。
3.在 「已学习模板」 下拉框中选择一条模板。
4.需要套用到当前图时，点击 「应用选中模板」。
5.不再需要某模板时，选中后点 「删除选中模板」。

# Design description

## 设计说明

### Analysis & Categorization
### 分析 · 归类入库

1. Ensure you have uploaded the poster and clicked "Confirm Current Grid" (saving without confirmation will usually prompt you to confirm first).

2. Enter the title in "Poster Name".

3. In "Tags", separate multiple tags with English or Chinese commas, such as: Event, E-commerce, Vertical.

4. Click "Save to Grid Library".

5. After successful saving, the corresponding card will appear in the list on the right; the "Grid Library" dropdown and "Apply Grid Library Grids" will also update.

1.确保已上传海报且已点击 「确认当前网格」（未确认时保存通常会提示先确认）。
2.在 「海报名称」 填写标题。
3.在 「标签」 用英文或中文逗号分隔多个标签，如：活动,电商,竖版。
4.点击 「保存到网格库」。
5.保存成功后，右侧列表会出现对应卡片；「网格库」下拉与「应用网格库网格」也会更新。

# Design description

## 设计说明

### Analysis & center page
### 分析 · 中心版面

1. Ensure a poster already exists and grid recognition is complete (consistent with the central canvas).

2. Click "Analyze Text/Image Areas" and wait for the status indicator to complete.

3. View the statistics in the `layoutAnalysisMeta` text block (number of text areas, number of image areas, etc.).

4. Set "Analysis Area Overlay" to "Display." You will see blue/red semi-transparent partitions on the canvas (if the imported layout is an editable layer, the overlay may automatically move aside).

5. When you need to access draggable layout, follow the prompts after analysis to enter layout editing mode; you can also continue adding text and image blocks in "Poster Layout."

1.确保当前已有海报且已完成网格识别（与中间画布一致）。
2.点击 「分析文字/图片区域」，等待状态提示完成。
3.查看 layoutAnalysisMeta 文本块中的统计（文字区数量、图片区数量等）。
4.将 「分析区域叠加」 设为 「显示」，可在画布上看到蓝/红半透明分区（若已导入排版为可编辑图层，叠加可能自动让位）。
5.需要进入可拖拽排版时，分析完成后按提示会转入排版编辑；也可在「海报排版」里继续添加文字块、图片块。

# Design description

## 设计说明

### Analysis & color
### 分析 · 色彩分析

1. Scroll down to "Poster Color Extraction" and click the title to expand.

2. Ensure a poster has been uploaded; the system will automatically display up to approximately 14 main color blocks.

3. Click on a color block to copy the HEX image to your clipboard (depending on your browser permissions).

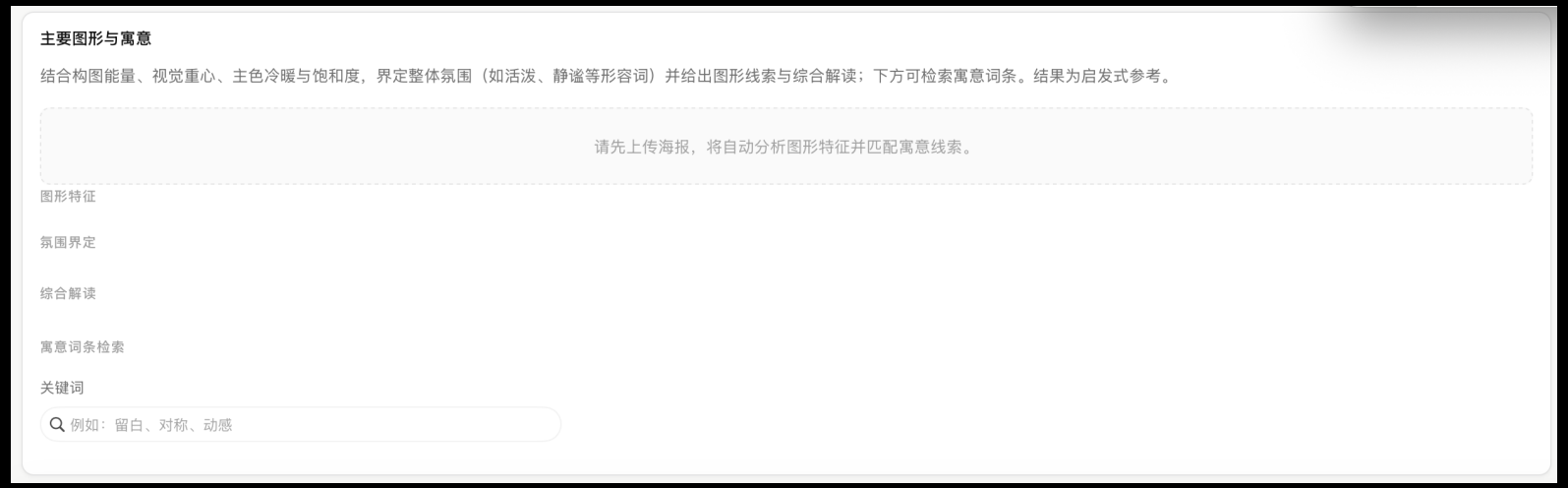
4. Next to each color block, you can view its RGB and converted CMYK reference values and approximate proportions.

1.向下滚动到 「海报颜色提取」， 点击标题展开。
2.确保当前已上传海报；系统自动显示最多约 14 个主色色块。
3.点击某一色块，将 HEX 复制到剪贴板（以浏览器权限为准）。
4.每个色块旁可查看 RGB 与换算的 CMYK 参考值及大致占比。

# Design description

## 设计说明

### Analysis & Meaning
### 分析 · 海报寓意




1. Expand "Main Graphics and Symbolism".

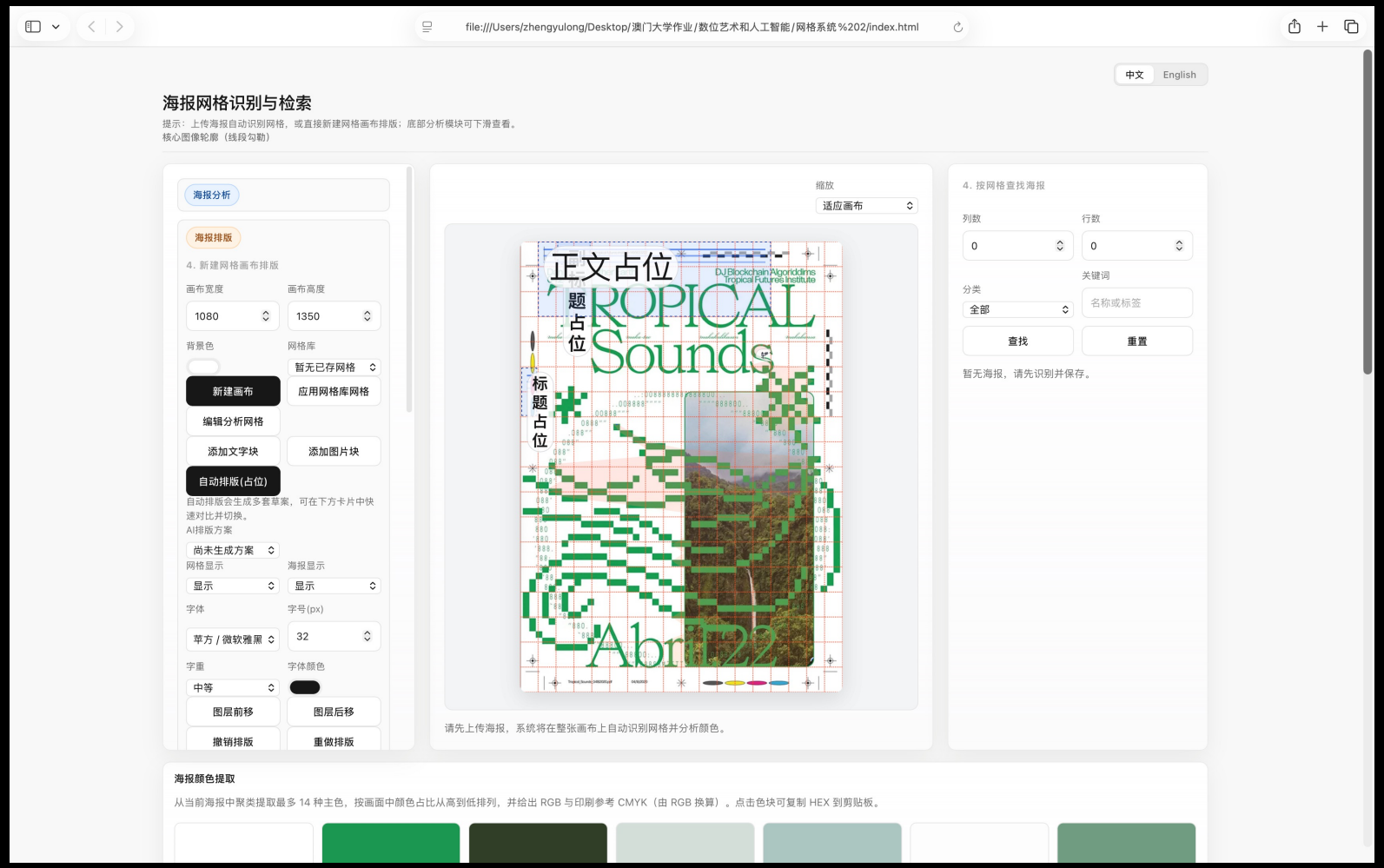
2. Read the automatically generated graphic features, atmosphere definition, and comprehensive interpretation (for reference only).

3. Enter keywords (such as "white space" or "symmetry") in the "Symbolic Meaning Search" field; the list below will filter and display the descriptions of the terms.

1.展开 「主要图形与寓意」。
2.阅读自动生成的 图形特征、氛围界定、综合解读（仅供参考）。
3.在 「寓意词条检索」 输入关键词（如「留白」「对称」），下方列表过滤显示词条说明。

# Design display

设计展示

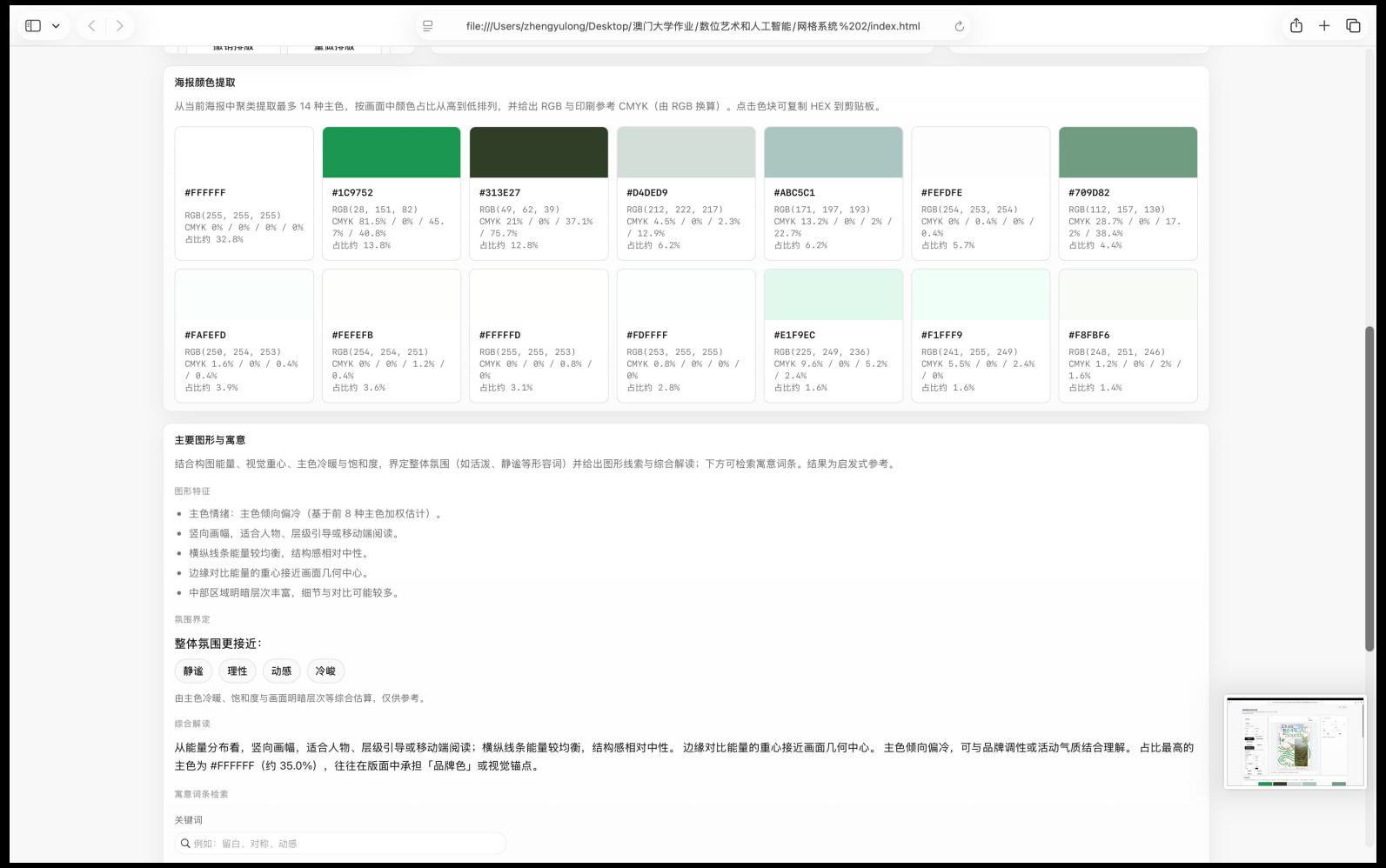



Poster Grid Analysis and Summary

海报网格分析与归纳

# Design display

设计展示




Color analysis and color value display

色彩分析与色值显示

# Design display

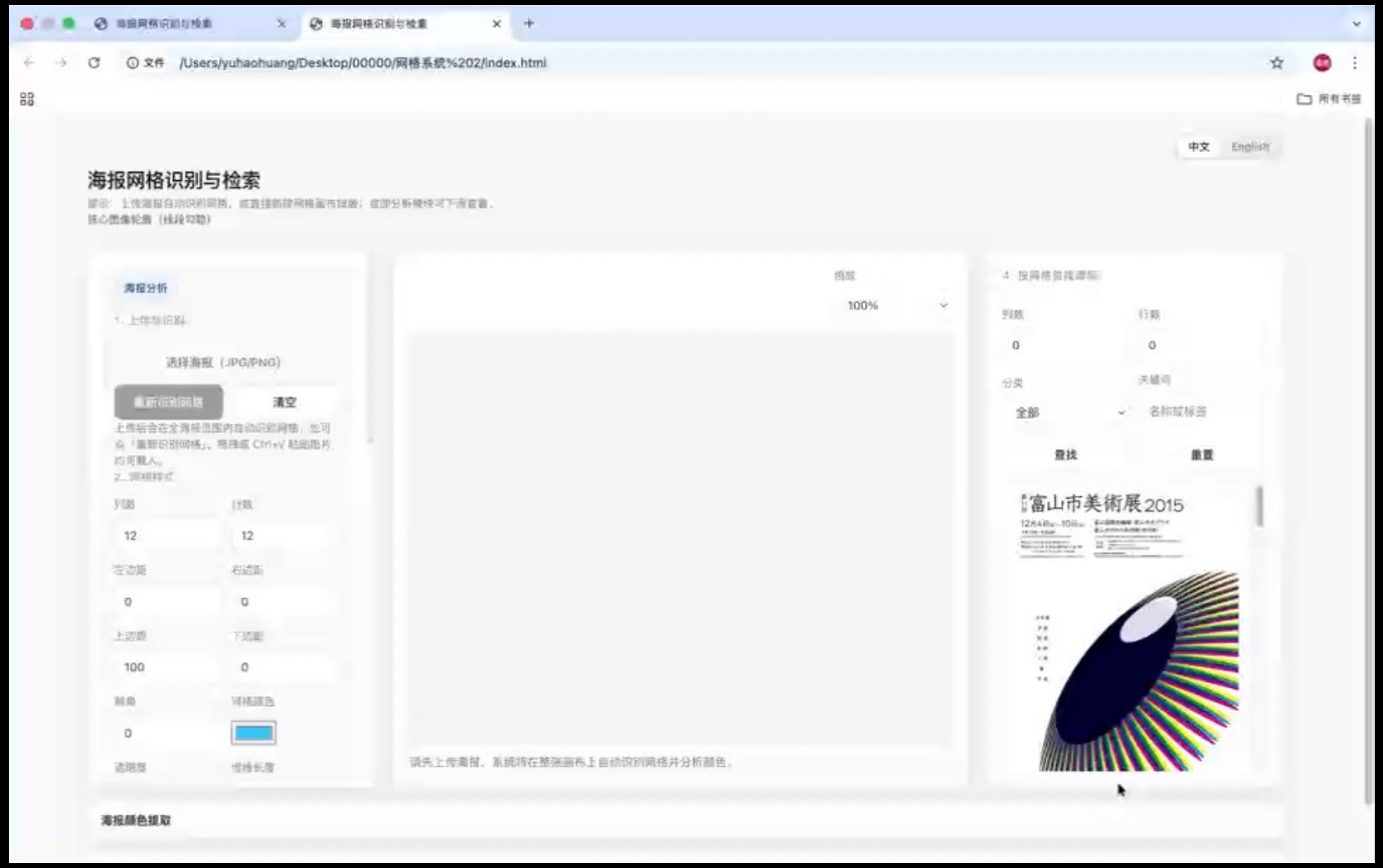
设计展示

Cultural connotation and atmosphere judgment

文化寓意和氛围判断

# Explainer video

## 视频讲解




https://youtu.be/0uA5z9_bn-A

## Design function

## 设计作用

This webpage allows for detailed analysis of poster grid systems. This can quickly improve the sensitivity of design beginners to poster observation. At the same time, it provides convenience for professional designers when organizing and optimizing their design work. In particular, the categorization and indexing features facilitate better analysis and organization of design works.

通过该网页，可以详细的对海报的网格系统进行分析。这可以很快的提高设计初学者对海报观察的敏感度。同时对于专业设计师来说对整理和优化自己的设计作品时，也提供了便利。尤其归类和索引，可以更好的对设计作品解析和整理。

Thanks for watching

# 感谢观看

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